MATH6660 - Finite Element Methods

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Context:

5610 : interp, int, ODEs	6660 : FEMs
5620 : num. lin alg.	6670 : FVMs
	5170 : FDMs

→ Why FEMs ?

(1) Theoretical:

FDM : approx function values

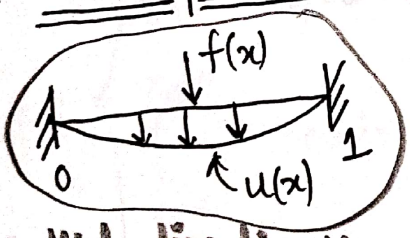
FEM : keep continous problem (PDEs), but replace (approx) function spaces. (so the soln inherits all the properties of PDEs).

(2) Practical reason:

- Can be formulated on unstructured grids, lots of flexibility in choice of function spaces. And this is also relatively simple to do.

Note: much of theory of FEMs is based on elliptic PDEs theory. Parabolic PDEs are OK. Hyperbolic PDEs are more challenging. This is where FEM starts to fail.

Two-point BVP : (1D version of Poisson).



Misleading figure. Try to think of it like a heat transfer problem, rather than a structural mechanics problem.

$$-(a(x)u')' = f(x) \qquad 0 < x < 1$$

$$u(0) = 0, \quad u(1) = 0$$

$u :=$ "displacement"

$f :=$ "external load" (given)

$a :=$ "modulus of elasticity" (given)

assume : $0 < a_1 \leq a(x) \leq a_2 < \infty \quad \forall \; x \in [0,1]$.

Note: $\mathcal{L}(u) := -(au')'$ with $u(0) = u(1) = 0$ is a linear differential operator that is self-adjoint.

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$$\int_0^1 v\mathcal{L}(u)\,dx = -\int_0^1 v(au')'dx$$

$$= \int_0^1 v'au'dx - [vau']_{x=0}^{x=1}$$

assume $u(0)=u(1)=0$ and $v(0)=v(1)=0$.

$$= \int_0^1 v'au'dx = -\int_0^1 u(av')'dx + [uav']_{x=0}^{x=1}$$

$$= -\int_0^1 u(av')'dx = \int_0^1 u\mathcal{L}(v)\,dx$$

$$\therefore \int_0^1 v\mathcal{L}(u)dx = \int_0^1 u\mathcal{L}(v)\,dx \Rightarrow$$ This is what it means to be self-adjoint.

Note: self-adjoint $\mathcal{L}$ is analogous to symmetric matrices.

Note: $\mathcal{L}(w)=0$, $w(0)=w(1)=0$ $\Leftrightarrow w\equiv 0$

* $\mathcal{L}(w)=0 \Rightarrow -(a(x)w')'=0 \Rightarrow w=\frac{c_1}{a}x+c_2$
  ∴ $w=0$ @ $x=0+x=1 \Rightarrow \boxed{w=0}$

* if $w=0 \Rightarrow \mathcal{L}(w)=-(a\cdot 0)'=0$.

$\mathcal{L}(w)$ is similar to Aw for some A. Since in linear algebra (discrete form) Ax is a linear operator acting on variable x. Similarly in continous form $\mathcal{L}(w)$ is a linear diff operator acting on w. So since $Aw=0 \Rightarrow w=0$, this means A is non-singular and is invertible.

∴ $\mathcal{L}$ is "invertible" ⇒ for a given a & f, there exists a unique u.

Note: if $0<a<\infty$ and f are sufficiently smooth. (need: $a\in C^1(0,1)$ and $f\in C^0(0,1)$) then $u\in C^2(0,1)$

Variational formulation:

Multiply BVP by a test function and integrate over $[0,1]$.

$$-\int_0^1 (au')'v\,dx = \int_0^1 fv\,dx.$$ then integrate-by-parts.

$$\Rightarrow \int_0^1 au'v'dx - [au'v]_0^1 = \int_0^1 fv\,dx$$

Assume v satisfies B.Cs: $v(0)=v(1)=0$.

The reason why we take $v=0$ at $x=0 \& 1$ is because v is δu. There is no other logical explanation.

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In particular, define the following space:

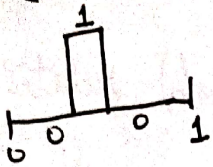
$$V = \left\{ v : \int_0^1 v^2 dx < \infty \,,\ \int_0^1 [v']^2 dx < \infty \,,\ v(0) = v(1) = 0 \right\}$$

→ So basically these two integrals just imply square continuity & finiteness over Ω.

[The squaring of $v' \times v'$ in these formulas is done because of some reason related to the use of L^2 norm].

Note: functions in V are continous ($V \in C^0(0,1)$)

(1) $\int_0^1 v^2 dx < \infty$ admits discontinous functions.



(2) $\int_0^1 [v']^2 dx$ does not allow discontinuities in V

* So taking the residual of the BVP and multiplying it by a test function and equating it equal to zero in Weighted residual method which is applicable for both self-adjoint as well as non-self adjoint problems.
• In variation calculus we say that variational formulation is where the energy functional of the system minimizes. The energy function or mathematically, the Lagrangian can only be formulated for self-adjoint problems and when you formulate it and then minimize it, the expression you get the so called variational formulation is same as the weighted residual statement of that BVP. That's why sometimes people call weighted residual statement as variational formulation for self-adjoint BVPs.
• But there are some engineering systems like problems with directional flow like fluid dynamics, heat convection, etc which are non-self adjoint and give non-symmetric stiffness matrices and hence people use Petrov-Galerkin for such problems and also different expensive solving process is used for such unsymmetric problems.
• Why "variational"?? $\Rightarrow J(v) = \frac{1}{2}\int_0^1 a(x) v'^2 dx - \int_0^1 fv\,dx$
$\frac{d}{d\varepsilon} J(u+\varepsilon v)\Big|_{\varepsilon=0} = 0$ v → arbitrary test functions that vanishes but the boundaries ($\delta u = v$)
$\Rightarrow \int a(x) u'v'dx - \int fv\,dx = 0$; Because test fnc's v' represent these small directional variations (δu) away from the exact solution u, the resulting integral equation is called the variational formulation.

ex:

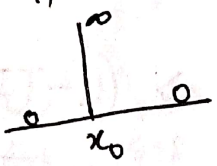
$$H(x) = \begin{cases} 1 & \frac{1}{4} < x < \frac{3}{4} \\ 0 & \text{otherwise} \end{cases}$$

$H \notin C^0(0,1)$, $H \in C^{-1}(0,1)$

$$\int_0^1 H^2 dx = \int_{1/4}^{3/4} dx = \frac{1}{2}$$

$$H'(x) = \delta\left(x - \tfrac{1}{4}\right) - \delta\left(x - \tfrac{3}{4}\right)$$

$\delta(x - x_0)$: ⊛



$$\int_{x_0-\varepsilon_1}^{x_0+\varepsilon_2} \delta(x - x_0) dx = 1.$$

for any $\varepsilon_1, \varepsilon_2 > 0$

$$\int_{x_0-\varepsilon_1}^{x_0+\varepsilon_2} f(x)\,\delta(x - x_0) dx = f(x_0)$$

$$\int_0^1 (H')^2 dx = \int_0^1 \left(\delta_{1/4} - \delta_{3/4}\right)^2 dx = \int_0^1 \left(\delta_{\frac{1}{4}}^2 + \delta_{\frac{3}{4}}^2 - 2\,\delta_{\frac{1}{4}}\delta_{\frac{3}{4}}\right) dx$$

So, $\int_0^1 \delta\left(x - \tfrac{1}{4}\right)\delta\left(x - \tfrac{1}{4}\right) dx = +\infty$

$\therefore H(x) \notin V$ because H' is not square integrable.

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⇒ Variational Problem:

Find $u \in V$ such that $\int_0^1 a u'v'dx = \int_0^1 fv\,dx$ for all $v \in V$.

Note: u is the trial function & v are the test functions.

In this case, u & v come from the same function space V.

Q: How are (2-pt BVP) and (Var Prob) related?

- BVP requires $u \in C^2(0,1)$
- Var Prob only need $u \in V$. which is just C^0.

[So weak form or variational form reduces the continuity requirement by 2. $C^2 \to C^0$]

Claim 1: If Var Prob has a soln, it is unique.

Proof: (by contradiction).

assume $\exists$ 2 different solns: $u \neq U$

$$\int_0^1 a u'v'dx = \int_0^1 fv\,dx \quad \& \quad \int_0^1 aU'v'dx = \int_0^1 fv\,dx$$

$$\Rightarrow \int_0^1 a(u-U)'v'dx = 0 \qquad \text{take } v = u-U \in V$$

$$\Rightarrow \int_0^1 a((u-U)')^2 dx = 0 \Rightarrow (u-U)' = 0 \text{ on } (0,1)$$

$$\Rightarrow u = U + c$$

but, by B.Cs : $c = 0 \Rightarrow u \in U$. ▨

Claim 2: Soln to BVP is a soln to Var. Prob.

Proof: BVP has a unique soln if $u \in C^2(0,1)$ and $u(0) = u(1) = 0$

If $u \in C^2(0,1)$ + B.Cs $\Rightarrow u \in V$

Then BVP & Var Prob are equivalent via int-by-parts. ▨

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Claim 3: if $f \in C^0(0,1)$, $a \in C^1(0,1)$, and $0 < a_1 \le a(x) \le a_2 < \infty$ $\forall x \in (0,1)$ then soln to Var. Prob. is a soln to BVP.

Proof: assume for the moment that $u \in C^2(0,1)$.

$$\int_0^1 (au'v' - fv)\,dx = 0$$

reverse int. by parts: $\int_0^1 \underbrace{[(au')' + f]}_{S(x)} v\,dx = 0$

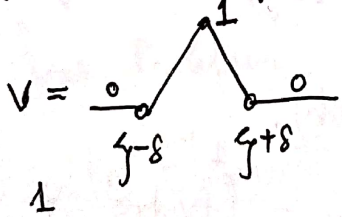
Goal: to show $S(x) \equiv 0$ for $x \in (0,1)$

(a) Assume $S(\xi) = [(au')' + f]\Big|_{x=\xi} > 0$ at some point $\xi \in (0,1)$.

Note that $S(x)$ is continous in x.

By continuity: $S(x) > 0$ for $x \in [\xi - \delta, \xi + \delta]$ for some $\delta > 0$.

Choose v to be the following "hat function".



but $\int_0^1 S(x)\,v(x)\,dx > 0$ which is a contradiction

$\therefore$ $S(x)$ can't be positive in $(0,1)$.

(b) Redo this argument w/ $S(\xi) < 0 \Rightarrow S(x)$ can't be -ve in $(0,1)$.

$\therefore$ $-(au')' = f$ on $(0,1)$.

$\Rightarrow$ When can we expect $u \in C^2(0,1)$?

(*) { if $f \in C^0(0,1)$, $a \in C^1(0,1)$, and $0 < a_1 \le a(x) \le a_2 < \infty$ $\forall x \in (0,1)$. }

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Note: (1) if (*) is satisfied: (2-pt BVP) $\equiv$ (Var. Prob.)
& u is a strong soln of (2-pt BVP)

(2) if (*) is not satisfied, (2-pt BVP) will not have a soln, but (Var. Prob.) may still have a soln.
$\Rightarrow$ u (if it exists) is a weak soln of (2-pt BVP)

"Energy" minimization problem:

Define the following "energy" functional:

$$F(w) = \underbrace{\frac{1}{2}\int_0^1 a\cdot(w')^2 dx}_{\text{internal energy}} - \underbrace{\int_0^1 f w\, dx}_{\text{load potential energy}}$$

Min problem: Find $u \in V$ such that $F(u) \leq F(w)$ $\forall\, w \in V$.

Claim: (Min problem) and (Var. Prob) are equivalent.

Proof: (1) (Var. Prob) $\Rightarrow$ (Min Prob):

Assume $u \in V$ satisfies: $\int_0^1 a u'v' dx = \int_0^1 fv\, dx \quad \forall\, v \in V$

Let $w \in V$, set $w = u+v$.

$$F(w) = F(u+v) = \frac{1}{2}\int_0^1 a(u'+v')^2 dx - \int_0^1 fu\, dx - \int_0^1 fv\, dx$$

$$= \frac{1}{2}\int_0^1 a(u')^2 dx + \int_0^1 au'v' dx + \frac{1}{2}\int_0^1 a(v')^2 dx - \int_0^1 fu\, dx - \int_0^1 fv\, dx$$

$$= F(u) + \frac{1}{2}\int_0^1 a(v')^2 dx.$$

$$\Rightarrow F(w) \geq F(u)$$

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$\therefore F(u) \le F(w) \quad \forall w \in V.$

(2) (Min. Prob) $\Rightarrow$ (Var. Prob):

Assume $u \in V$ satisfies $F(u) \le F(w) \ \forall w \in V$

Let $g(\varepsilon) = F(u+\varepsilon v)$ where $v \in V$ is arbitrary, but fixed.

$$g(\varepsilon) = \frac{1}{2}\int_0^1 a(u'+\varepsilon v')^2 dx - \int_0^1 f\cdot(u+\varepsilon v)dx$$

$$= \frac{1}{2}\int_0^1 a(u')^2 dx + \varepsilon\int_0^1 au'v'dx + \frac{1}{2}\varepsilon^2\int_0^1 a(v')^2 dx - \int_0^1 fu\,dx - \varepsilon\int_0^1 fv\,dx$$

$g(\varepsilon)$ is a quadratic poly in ε w/ a positive quad. term.

$\therefore$ the unique global minimizer happens at $\varepsilon = 0$ and $g'(\varepsilon=0)=0$.

$$g'(\varepsilon) = \int_0^1 au'v'dx - \int_0^1 fv\,dx + \varepsilon\int_0^1 a(v')^2 dx$$

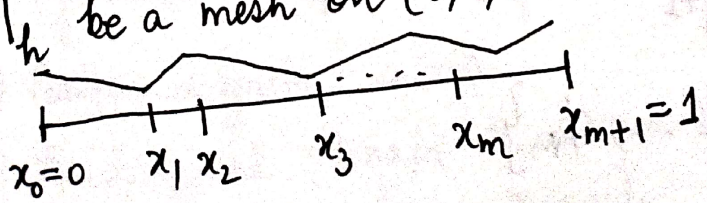
$$g'(\varepsilon=0) = \int_0^1 au'v'dx - \int_0^1 fv\,dx = 0, \text{ which is exactly the}$$

Var. Problem. ∎

$\Rightarrow$ Our 1st Finite Element Method:

Idea: Solve (Var. Prob) or (Min. Prob) approximately by replacing V by a finite dimensional subspace V_h.

Let T_h be a mesh on $(0,1)$:



Let $h_j := x_j - x_{j-1}$ for $j = 1, 2, \ldots, M+1$.

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Simplest $V_h \subset V$: space of continous piecewise linear polynomials on T_h.

This is called the cG(1) method.

continous Galerkin method w/ polys of degree 1.

Any function in V_h can be uniquely ~~within~~ written as :

$$U(x) = \sum_{j=1}^{M} \xi_j \phi_j(x)$$ where $\phi_j(x)$ is a "hat function" associated to x_j.

$$\phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{h_j} & x_{j-1} \le x \le x_j \\ \frac{x_{j+1} - x}{h_{j+1}} & x_j \le x \le x_{j+1} \end{cases}$$

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T_h : mesh w/ grid pts. $x_0 = 0, x_1, x_2, \cdots, x_m, x_{m+1} = 1$

FEM space : $V_h := \{ v \in V : v|_{[x_i, x_{i+1}]} \in$ Polynomial of degree 1 $\forall\ i = 0, \cdots, M \}$

FEM : Find $U \in V_h \subset V$ s.t $\int_0^1 a u' v' dx = \int_0^1 f v dx \quad \forall\ v \in V_h$.

A basis for V_h : $\phi_1(x), \phi_2(x), \cdots, \phi_m(x)$

where $\phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{h_j} & x_{j-1} \le x \le x_j \\ \frac{x_{j+1} - x}{h_{j+1}} & x_j \le x \le x_{j+1} \end{cases}$

$V_h = \text{Span}\{\phi_1, \phi_2, \cdots, \phi_M\}$

$\dim V_h = M$.

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if $U \in V_h$, then $U(x) = \sum_{j=1}^{M} \xi_j \phi_j(x)$ where $\xi_j = U(x_j)$ for $j = 1, 2, \ldots, M$.

Galerkin Orthogonality:

$$\int_0^1 a u' v' dx = \int_0^1 f v dx, \quad u, v \in V$$

$$\int_0^1 a U' v' dx = \int_0^1 f v dx, \quad U, v \in V_h$$

Subtract: $\int_0^1 a(\overbrace{u - U}^{\text{error}})' v' dx = 0$ for all $v \in V_h$.
error

Let $U(x) = \sum_{j=1}^{M} \xi_j \phi_j(x)$ $\quad v = \phi_i \quad i = 1, 2, \ldots, M.$

$$\Rightarrow \int_0^1 a\left(\sum_{j=1}^{M} \xi_j \phi_j\right)' \phi_i' dx = \int_0^1 f \phi_i dx$$

$$\Rightarrow \sum_{j=1}^{M} \xi_j \int_0^1 a \phi_i' \phi_j' dx = \int_0^1 f \phi_i dx$$

This is a ~~linear~~ $M \times M$ linear system:

$$\underline{\underline{A}}\,\underline{\xi} = \underline{b} \quad : \quad A_{ij} = \int_0^1 a \phi_i' \phi_j' dx;$$

$$, b_i = \int_0^1 f \phi_i dx; \; \underline{\xi} = (\xi_1, \xi_2, \ldots, \xi_M).$$

$\underline{\underline{A}}$: stiffness matrix

(1) Symmetric : test & trial functions are from the same space.

(2) Positive definite : $a > 0$.

(3) tridiagonal : basis functions have compact support on $[x_{i-1}, x_{i+1}]$.

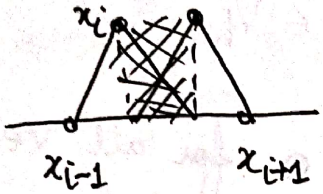
$$\int_0^1 a \phi_i' \phi_j' dx \neq 0 \quad \text{for } j = i-1, i, i+1.$$

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ex: $a \equiv 1$ and take $h_j \equiv h \ \forall j$. (uniform mesh).

$$A_{ii} = \int_0^1 \phi_i' \phi_i' dx = \int_{x_{i-1}}^{x_i} (\phi_i')^2 dx + \int_{x_i}^{x_{i+1}} (\phi_i')^2 dx = \frac{1}{h^2}(x_i - x_{i-1}) + \frac{1}{h^2}(x_{i+1} - x_i)$$
$$= \frac{2}{h}$$

$$A_{(i,i+1)} = \int_{x_i}^{x_{i+1}} \phi_i' \phi_{i+1}' dx = -\frac{1}{h}$$



$$b_i = \int_{x_{i-1}}^{x_{i+1}} f(x) \phi_i(x) dx \overset{\text{Composite trapezoidal rule}}{\approx} \frac{h}{2}\Big[f(x_{i-1}) \cdot \overbrace{\phi_i(x_{i-1})}^{0} + 2f(x_i)\phi_i(x_i) + f(x_{i+1})\overbrace{\phi_i(x_{i+1})}^{0}\Big]$$
$$= h f(x_i)$$

$$\underline{\underline{A}} = \frac{1}{h}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & & -1 & \\ & & & -1 & 2 \end{bmatrix}$$

$$\underline{b} \approx h\,(f(x_1), f(x_2), \ldots, f(x_m))$$

Note: this is exactly $O(h^2)$ central finite difference method.

2/9/26

Nonhomogeneous Dirichlet BCs:

$$\begin{cases} -(au')' = f \quad \text{in } (0,1) \\ u(0) = \alpha \ \text{&} \ u(1) = \beta \end{cases}$$

Var. prob:

$$\int_0^1 au'v'dx - [au'v]_0^1 = \int_0^1 fv\,dx$$

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To make boundary term vanish:

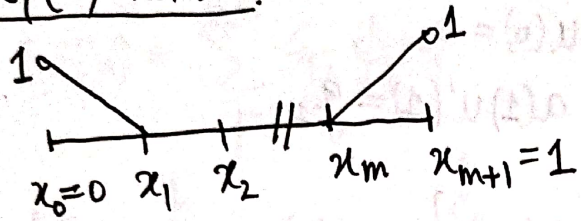
$$V = \{ v : \int_0^1 [v^2 + (v')^2]\,dx < \infty \;,\; v(0) = \alpha \;\&\; v(1) = \beta \}$$

$$V_0 = \{ v : \int_0^1 [v^2 + (v')^2]\,dx < \infty \;,\; v(0) = v(1) = 0 \}$$

Var. Prob:

Find $u \in V$ s.t $\int_0^1 a u' v' dx = \int_0^1 f v\,dx \quad \forall\, v \in V_0$.

CG(1) method:



use same basis : $\phi_j(x)$ • $j = 1, \cdots, M$
& introduce 2 "half-hat" functions*:

$$\phi_0(x) = \frac{x_1 - x}{x_1 - x_0} \qquad x \in [x_0, x_1]$$

$$\phi_{M+1}(x) = \frac{x - x_m}{x_{m+1} - x_m} \qquad x \in [x_M, x_{M+1}]$$

Trial function:

$$U(x) = \alpha \phi_0(x) + \beta \phi_{M+1}(x) + \sum_{j=1}^{M} \xi_j \phi_j(x)$$

Test functions:

$$V = \phi_i(x) \qquad i = 1, \cdots, M.$$

$$\Rightarrow \int_0^1 a U' v' dx = \int_0^1 f v\,dx$$

$$\Rightarrow \sum_{j=1}^{M} \xi_j \int_0^1 a \phi_i' \phi_j' dx + \alpha \int_0^1 a \phi_0' \phi_i' dx + \beta \int_0^1 a \phi_{M+1}' \phi_i' dx = \int_0^1 f \phi_i dx$$

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$$\Rightarrow \sum_{j=1}^{M} c_j \int_0^1 a\phi_i' \phi_j' dx = \int_0^1 f\phi_i dx + \delta_{i1}\frac{\alpha}{h_1^2}\int_{x_0}^{x_1} a(x)dx + \delta_{Mi}\frac{\beta}{h_{M+1}^2}\int_{x_m}^{x_{M+1}} a(x)dx$$

Note: • Dirichlet B.Cs are enforced **strongly** and modify the RHS.

• In FEM language, Dirichlet B.Cs for 2-pt BVPs are called **essential B.Cs.** (i.e, must be imposed in the definition of the function space V).

$\Rightarrow$ **Neumann B.Cs:**

$$\begin{cases} -(au')' = f & \text{in } (0,1) \\ u(0) = 0 \\ a(1)u'(1) = g_1 \end{cases}$$

$$V = \left\{ v : \int_0^1 (v^2 + (v')^2)dx < \infty,\ v(0) = 0 \right\}$$

Int-by-parts:

$$\int_0^1 au'v'dx - \underbrace{a(1)u'(1)}_{g_1}v(1) + a(0)u'(0)\cancel{v(0)}^{0} = \int_0^1 fv\,dx$$

$$\Rightarrow \int_0^1 au'v'dx - g_1 v(1) = \int_0^1 fv\,dx \quad \forall\ v \in V.$$

Q: Is soln to Var. prob. also a soln to 2-pt BVP?

$\Rightarrow$ Start with Var. Prob & reverse int-by-parts:

$$-\int_0^1 (au')'v\,dx + a(1)u'(1)v(1) = \int_0^1 fv\,dx + g_1 v(1)$$

This must hold for all $v \in V$, including all $v \in V$ w/ $v(1) = 0$.

$$\int_0^1 fv\,dx = -\int_0^1 (au')'v\,dx \quad \forall\ v \in V \cap v(1) = 0.$$

$\Rightarrow -(au')' = f$ in $(0,1)$

$\therefore a(1)u'(1)v(1) = g_1 v(1) \Rightarrow a(1)u'(1) = g_1.$

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cG(1) FEM:

$$V_h = \{ v \in V : v|_{[x_j, x_{j+1}]} = \text{poly of degree } 1. \}$$

$$V_h = \text{Span}\{\phi_1, \phi_2, \ldots, \phi_M, \phi_{M+1}\}$$

(ϕ_{M+1}: new part)

$\phi_{M+1}(x)$ is the half-hat function at right end point

Trial function:

$$U(x) = \sum_{j=1}^{M+1} \xi_j \phi_j(x)$$

s.t. $$\int_0^1 a U' v' dx = \int_0^1 f v dx + g_1 v(1). \quad \forall v \in V_h.$$

$v = \phi_i \quad i = 1, 2, \ldots, M, M+1$ (new part)

$\phi_i(1) = 0 \quad \forall i = 1, 2, \ldots, M$

$\phi_{M+1}(1) = 1$ ← only one that is not zero at $x=1$.

4/9/26

$$\begin{cases} -(au')' = f \text{ in } (0,1) \\ u(0) = 0, \quad a(1)u'(1) = g_1 \end{cases}$$

$$V = \{ v : \int_0^1 [v^2 + v'^2] dx < \infty, \ \underbrace{v(0) = 0}_{\text{Strong}} \}$$

cG(1) method:

$$U(x) = \sum_{j=1}^{M+1} \xi_j \phi_j(x)$$

$$\int_0^1 a U' v' dx = \int_0^1 f v dx + \boxed{g_1 v(1)}_{\text{weak}}$$

$v = \phi_i \quad i = 1, 2, \ldots, M+1.$

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ex: $a=1$, $h_i \equiv h$ (uniform mesh).

$$\sum_{j=1}^{M+1} c_j \int_0^1 \phi_i' \phi_j' dx = \int_0^1 f\phi_i dx + g_1 \phi_i(0).$$

$$A_{(M,M+1)} = \int_{x_M}^{x_{M+1}} \phi'_{M+1}\, \phi'_M dx = -\frac{1}{h^2} \int_{x_M}^{x_{M+1}} dx = -\frac{1}{h} = A_{(M+1,M)}$$

$$A_{(M+1,M+1)} = \int_{x_M}^{x_{M+1}} (\phi'_{M+1})^2 dx = \frac{1}{h}$$

$$\underline{\underline{A}}\,\underline{c} = \underline{b} \qquad \underline{c} = (c_1, c_2, \cdots, c_{M+1})$$

$$\underline{\underline{A}} = \frac{1}{h}\begin{bmatrix} 2 & -1 & & & & \\ -1 & 2 & -1 & & & \\ & \ddots & \ddots & \ddots & & \\ & & -1 & 2 & -1 & \\ & & & -1 & 1 & \end{bmatrix}$$

$\underline{\underline{A}} \in \mathbb{R}^{(M+1)\times(M+1)}$, symm, pos. def.

$$b_i = \int_{x_{i-1}}^{x_{i+1}} f\phi_i dx \quad i = 1, \cdots, M, \qquad b_{M+1} = \int_{x_M}^{x_{M+1}} f\phi_{M+1} dx + g_1$$

Note: for 2-pt BVP
(1) Dirichlet BCs are strongly enforced. (essential)
(2) Neumann or mixed BCs are weakly enforced (natural).

⇒ Error Analysis:

Two kinds of error estimates:
(1) a priori: error estimate in terms of derivatives of exact soln.

e.g: $\|u - U\| \le c h^p \|u^{(l)}\|$

l^{th} derivative of u.

u: exact soln U: fem soln.

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(2) a posteriori : error estimates in terms of a "residual". (eg, $\mathcal{L}(U)-f$). Useful for calculating errors during the simulation ⇒ adaptive error control (adaptive mesh refinement).

Let $e(x) := u(x) - U(x)$ be the error.

"Energy" norm : $\|v\|_E := \sqrt{\int_0^1 a(x)\,[v'(x)]^2\,dx}$

Weighted L_2-norm : $\|w\|_a := \sqrt{\int_0^1 a(x)\,[w(x)]^2\,dx}$

Note: $\|v\|_E = \|v'\|_a$

⇒ Cauchy inequalities:

- $\left|\int_0^1 av'w'\,dx\right| \le \|v'\|_a \cdot \|w'\|_a$
  $= \|v\|_E \|w\|_E$

- $\left|\int_0^1 vw\,dx\right| \le \|v\|_a \|w\|_{a^{-1}}$

⇒ An a priori estimate:

cG(1) for Homogeneous Dirichlet B.Cs

$$\begin{cases} -(au')' = f \text{ in } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

$$\int_0^1 aU'v'dx = \int_0^1 fv\,dx \quad \forall\, v \in V_h \subset V$$

$$\int_0^1 au'v'dx = \int_0^1 fv\,dx \quad \forall\, v \in V$$

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$$\therefore \int_0^1 a\overbrace{(u-U)}^{e}v'dx = 0 \qquad \forall\ v \in V_h$$

↳ Galerkin Orthogonality.

⇒ **Energy norm a priori error:**

$$\|u-U\|_E^2 = \|(u-U)'\|_a^2 = \int_0^1 a(u-U)'(u-U)'dx \quad \text{for any } v \in V_h.$$

For any $v \in V_h$:-

$$\|u-U\|_E^2 = \int_0^1 a(u-U)'(u-v+v-U)'dx$$

$$= \int_0^1 a(u-U)'(u-v)'dx + \int_0^1 a(u-U)'(v-U)'dx$$

$\int_0^1 a(u-U)'(v-U)'dx \to 0$ by Galerkin orthogonality.

Note: $u-v \in V$, but $v-U \in V_h$

So, $\|u-U\|_E^2 = \int_0^1 a(u-U)'(u-v)'dx$ for any $v \in V_h$

using Cauchy:

$$\Rightarrow \int_0^1 a(u-U)'(u-v)'dx \leq \|(u-U)'\|_a \|(v-u)'\|_a$$

$$\Rightarrow \|u-U\|_E^2 \leq \|(u-U)'\|_a \cdot \|(u-v)'\|_a$$

$$\|u-U\|_E^2 \leq \|u-U\|_E \|u-v\|_E$$

$$\therefore \boxed{\|u-U\|_E \leq \|u-v\|_E} \quad \forall\ v \in V_h$$

∴ FEM soln U is the best soln (smallest energy norm).

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